

<https://github.com/surabhi84ard/Mini/tree>

Experiment: K-Map Based Logic Gate Design using VAMAN Board

Course: Digital Logic Design using Verilog HDL

Objective

To implement a simplified Boolean function derived from its Karnaugh Map (K-Map) and verify it on the VAMAN FPGA development board.

Apparatus Required

- VAMAN FPGA Development Board (Spartan-6)
- USB JTAG Programmer
- On-board switches and LEDs
- Xilinx ISE / Vivado Software

Boolean Function

The logic expression obtained from K-map simplification is:

$$f = (b + c)(a + \bar{c})$$

This expression represents the minimized Product of Sums (PoS) form.

Truth Table

A	B	C	F
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

K-Map Simplification

$AB \backslash C$	0	1
00	0	0
01	1	0
11	1	0
10	0	1

The grouped 1s in the K-map lead to:

$$f = (b + c)(a + \bar{c})$$

Verilog Code

```
1 module kmap_gate_vaman(  
2     input wire a,  
3     input wire b,  
4     input wire c,  
5     output wire f  
6 );  
7  
8 assign f = (b | c) & (a | ~c);  
9  
10 endmodule
```

Listing 1: Verilog Code for K-Map Simplified Logic Function

Pin Configuration (VAMAN Board Example)

Signal	FPGA Pin	Description
a	SW0	Input Switch 0
b	SW1	Input Switch 1
c	SW2	Input Switch 2
f	LED0	Output LED 0

Result

The output LED glows whenever the logic expression $(b + c)(a + \bar{c})$ evaluates to HIGH, confirming correct implementation of the simplified K-map expression.

Conclusion

The Verilog module successfully implements the minimized Boolean function obtained from the K-map. This verifies the use of logical reduction techniques for efficient hardware implementation on the VAMAN FPGA board.